

2.156 Challenge Problem 1: Linkage Synthesis

Screening, gradient refinement and growth of planar linkages

Prad R. [Teammate name]

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Abstract

We design planar linkages whose traced curve matches three kangaroo outlines while using as little link material as possible. Our pipeline is a mass random screen of about 21 million mechanisms, gradient refinement of the best of them, a growth stage that adds one joint at a time, a second, much wider refinement, and a second round of growth. The final submission scores **3.69** with the course scorer (hypervolumes 5.87, 8.52 and 24.60), up from 0.045 for the advanced starter notebook. Most of the score comes from the screen and the first refinement; after that, moving joints stopped helping and only changes to the link layout still lowered the distance. The tail of Kangaroo 3 is the main feature we do not capture.

1 Problem

A mechanism is a set of joints with initial positions, rigid links between them, a set of grounded joints, one motor link, and one traced joint. Turning the motor through a full revolution makes the traced joint draw a closed curve. For each of three target curves we submit up to 1000 mechanisms of at most 20 joints, and each mechanism is judged on two objectives, both minimized:

- **Distance:** the mismatch between the traced curve and the target after the best rotation and starting point. Scale is *not* normalized, so the curve has to be the right size as well as the right shape.
- **Material:** the total length of all links.

A design counts only if it is strictly under both limits in Table 1. The score for a target is the hypervolume its designs dominate relative to those limits, divided by a normalizer; the overall score is the mean over the three targets.

Problem	Target	Distance limit	Material limit	Normalizer
1	Kangaroo 1 (round body)	0.75	10	2.0
2	Kangaroo 2 (no ears or tail)	1.2	10	1.5
3	Kangaroo 3 (full outline)	1.75	20	10.0

Table 1: The three problems. The limits are both the validity cutoff and the hypervolume reference point.

Two properties of this setup shaped everything we did. First, scaling all joint positions by a factor scales the traced curve and the material together, so every mechanism has a free one-variable trade-off between the two objectives. Second, the hypervolume a design adds is its distance gain times the material headroom above it, so an accuracy gain on a cheap design is worth more than the same gain on a heavy one.

2 Approach

Table 2 lists the stages in the order they ran and the official score after each. All scores in this report come from the course function `evaluate_submission`.

Stage	What it does	Score after
Starter notebook	Seeded NSGA-II, then gradient descent (Kangaroo 2 only)	0.045
1. Screening	Random mechanisms, pruned and resized, no optimization	2.78
2. Refinement	Adam on joint positions, four material weights per design	3.43
3. Growth	Add one traced joint with two links, refine, select; 20 generations	3.50
4. Wide refinement	About 4000 starts per target, successive halving	3.67
5. Growth, round two	Growth again from the wide results, inside the limits	3.69

Table 2: Pipeline stages and the overall score after each.

2.1 Stage 1: mass random screening

We started by exploring instead of optimizing. The starter’s genetic algorithm found no feasible design from a random start and reached a score of only 0.045 when seeded, which suggested that the hard part was finding link layouts worth optimizing at all. Random sampling also parallelizes trivially across cluster workers, which a single evolving population does not.

Each worker repeatedly samples random mechanisms with 10 to 20 joints, built so that every joint hangs off two earlier ones, with several random joint layouts per link structure. All of them are solved in one batched call. We then treat *every* moving joint as a possible traced joint and prune its mechanism to the joints that actually drive it, which removes links that only add material. After pruning, the candidates have 4 to 9 joints.

Scoring every candidate with the official metric would be too slow, so the candidates are first ranked by a cheap shape comparison (a Fourier-domain least-squares alignment over all starting points, both directions, rotation and scale). The best 40 per target in each batch are then scored with the official metric at nine sizes around the best-fit scale, which turns each candidate into several points along its own distance and material trade-off.

Four workers ran for about four hours each. In total they sampled 20.9 million mechanisms, screened 91.8 million candidate curves, and made 22 million official evaluations. With no optimization at all this gave a score of 2.78 (Figure 1). The gains flattened early: the score was already 2.61 after the first worker-hour.

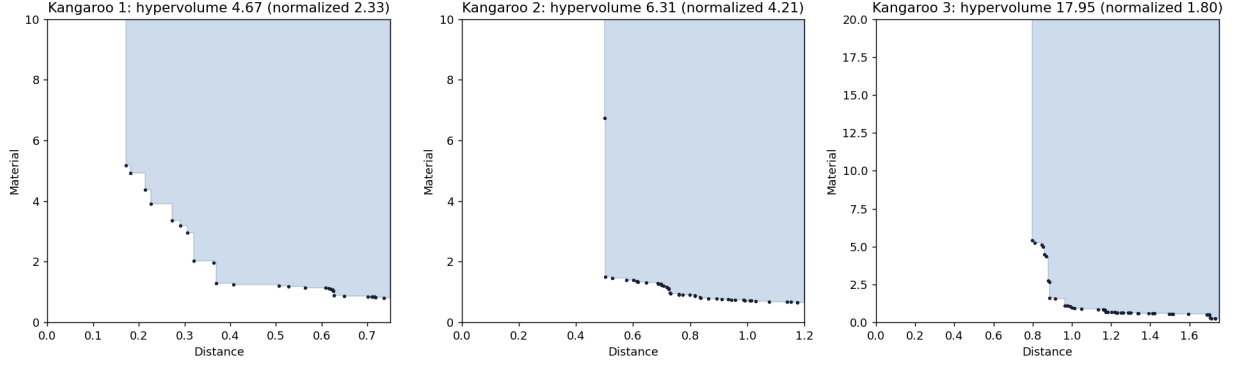


Figure 1: Pareto fronts after screening alone (score 2.78). Shaded area is the hypervolume. Best distances are 0.17, 0.50 and 0.80.

2.2 Stage 2: gradient refinement

For each target we took the most accurate screened designs, a fixed number per joint count so that larger mechanisms were represented (136 designs per target), and ran Adam on the joint positions for 1500 steps. Each design was refined four times, minimizing

$$\text{distance} + w \cdot \text{material}, \quad w \in \{0, 0.03, 0.1, 0.3\},$$

so that one starting design spreads into several points along the front. Every intermediate step that fell inside the limits was offered to the Pareto front, not only the final one.

This raised the score from 2.78 to 3.43 and cut the best distances to 0.067, 0.30 and 0.48 (Figure 2, blue). At this point the most accurate Kangaroo 3 design followed both ears, but the tail and feet were still missed, as were the snout and feet corners of Kangaroo 2.

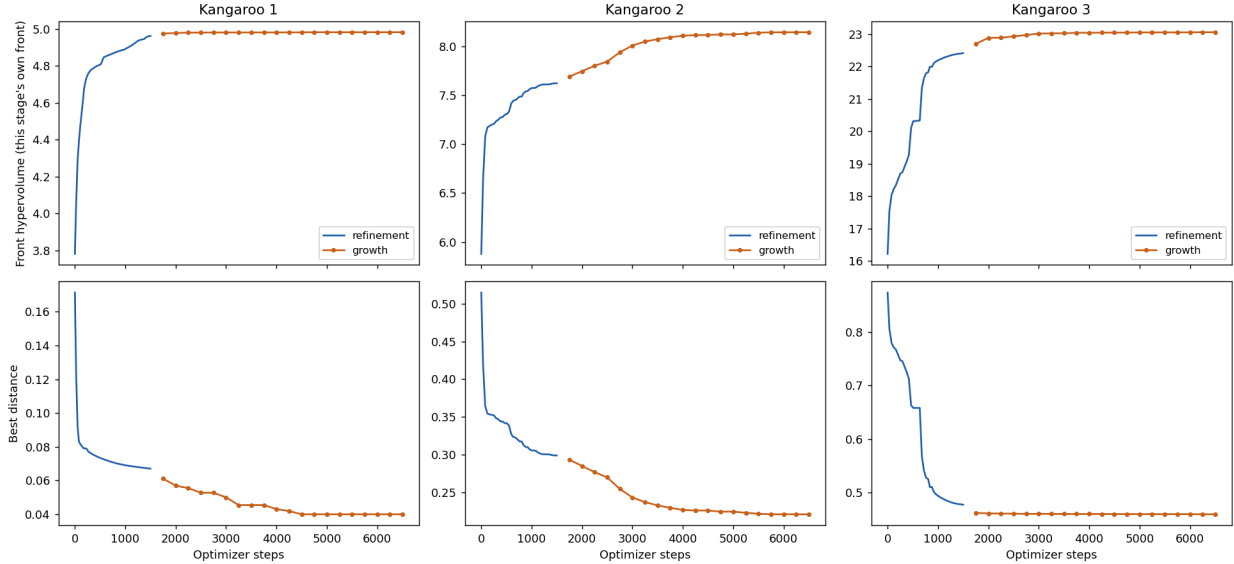


Figure 2: Hypervolume (top) and best distance (bottom) during refinement (blue) and growth round one (orange, one marker per generation). Each curve is that stage's own front, not the merged submission. In round one, growth selected parents on distance alone, so the Kangaroo 1 curve ends at a 20-joint design with distance 0.040 that is over the material limit and does not count; the best valid Kangaroo 1 distance after this stage was 0.054.

2.3 Stage 3: growth

Refinement only moves joints; it cannot change which joints are connected. To capture finer features we let designs grow. Each generation, every parent gets 96 children, each made by attaching *one new traced joint with two links* to the parent in one of three ways:

- to the old traced joint and one of its parents, which puts the new point on the same rigid body, close to the old one, so the child starts out tracing almost the same curve;
- to any two existing joints;
- to one moving joint and a new ground joint.

Children are pruned, resized to the target and scored. The best 512 are refined for 250 Adam steps, and the next parents are the most accurate designs in each joint-count bin, so small and large designs both survive.

Twenty generations raised the score to 3.50 and lowered the best distances to 0.054, 0.22 and 0.46. The largest effect was on Kangaroo 2 (0.30 to 0.22). The last generation found no improving child on any target.

One flaw surfaced here: parents were chosen on distance alone, so some drifted over the material limit and their gains did not count. We changed the selection to keep parents inside the limits before the second round.

2.4 Stage 4: wide refinement

The first refinement had started from only 136 designs per target, and its best results came from a few that happened to sit in good basins. It also contained no four-joint designs, although those hold the low-material end of every front. The wide round therefore restarts refinement from about 4000 candidates per target: all screening seeds and fronts, the results of stages 2 and 3, and the other two targets' best designs (the three outlines are versions of one shape). No new random mechanisms were sampled. Steps are spent by successive halving: a short run for all 4096, a longer one for the best 1024 (chosen per joint count), and a long run at several material weights for the best 192 plus 64 cheap designs.

This raised the score from 3.50 to 3.67. Notably, it did *not* improve the best distance on any target. The whole gain came from cheaper designs at the same accuracy, which is where hypervolume is most valuable.

2.5 Stage 5: growth, round two

Growth was run again from the wide results, with parents restricted to the material limits. It made the last improvements on all three problems and brought the score to 3.69. Every design set in the final file comes from this stage.

3 Where the remaining error is

Since the best distances had stopped moving, we measured the error point by point along each target, aligning designs exactly as the scorer does (Figure 3).

- **Kangaroo 1** is matched closely everywhere.

- **Kangaroo 2** has no single problem area. Its most accurate design has several similar peaks, at the snout tip, the feet and the lower-right corner. Cheap six-joint designs miss the concavity between snout and feet.
- **Kangaroo 3** loses most on the tail, between 55 and 65% of the perimeter. For the most accurate design the worst fifth of the perimeter carries 48% of the distance. The ears are captured.

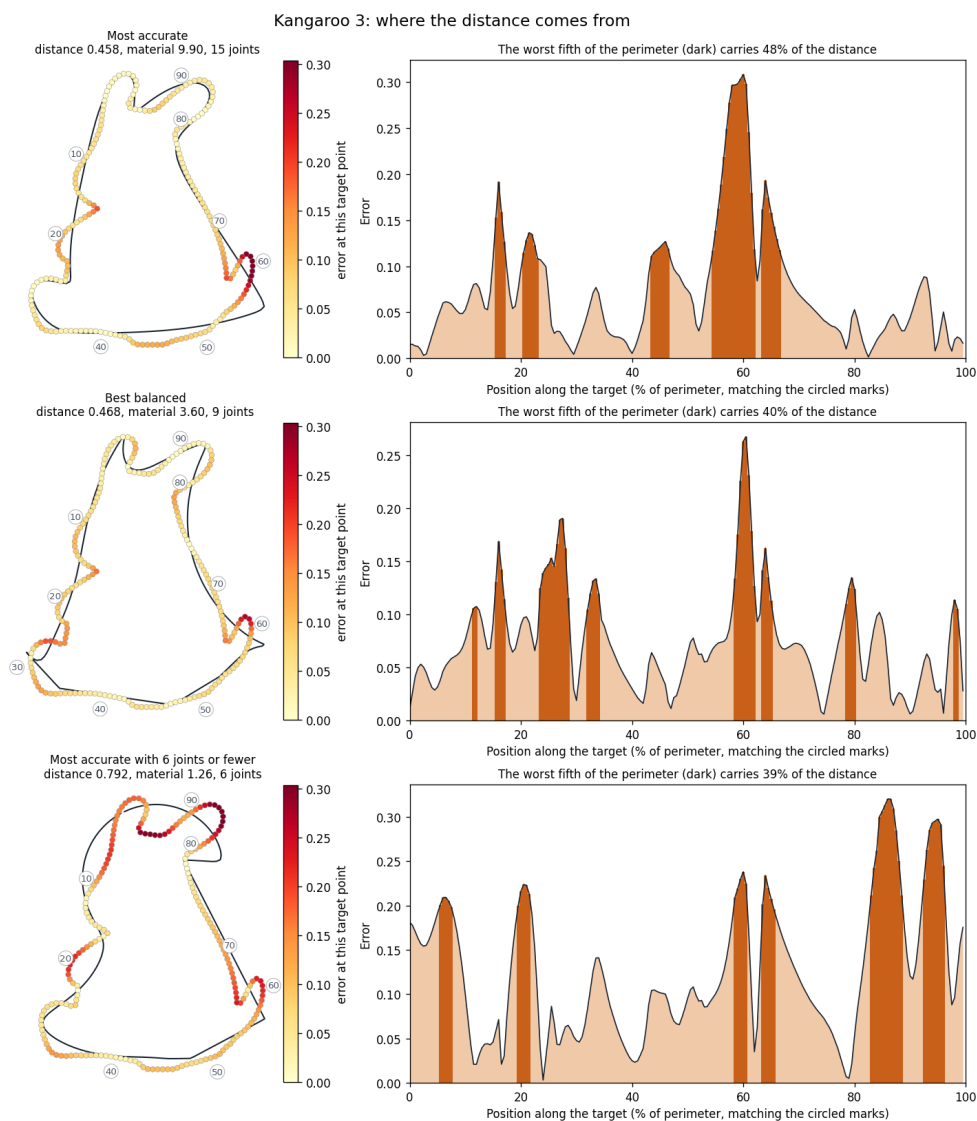


Figure 3: Kangaroo 3: error at each target point (left) and along the perimeter (right) for three designs, as of the evening of October 5. The tail, near 60% of the perimeter, dominates the most accurate design's distance.

The metric compares points at matching positions along the perimeter, so a curve that reaches part of the way into a narrow feature such as the tail earns little for it, and the optimizer has no easy path toward tracing it.

3.1 Ideas that did not help

We tested three ways of lowering distance without changing the link layout. All were negative.

Idea	Test	Hypervolume change (K1 / K2 / K3)
Assembly-branch flips	Mirror a moving joint across the line through its two parents, keeping every link length, then refine. About 7000 flips.	+0.012 / +0.001 / +0.001
More forgiving losses	Refine first on a two-way nearest-point distance, or on a blurred target sharpened in steps, then finish on the official distance. Compared with a control on the same parents.	Control matched or beat both on every target
Strictly improving polish	Accept only steps that improve one objective without worsening the other.	+0.0007 / +0.0005 / +0.0012

Table 3: Experiments on joint positions only.

The forgiving-loss test was our attempt to act on the tail diagnosis: if the official distance gives little credit for a partly formed tail, a gentler loss might lead the optimizer there. It did not. For Kangaroo 3 the control reached a hypervolume of 24.36 against 24.27 and 24.19 for the two variants. A side finding was that restarting Adam on converged designs usually left them worse than they began, so the best point along the way has to be kept.

Together with the wide refinement, these results say the front designs sit at true minima for their link layouts. Only structural change, the growth stage, still lowered distance.

4 Final results

Target	Designs	Hypervolume	Normalized	Best distance	Least material	Joints
Kangaroo 1	1000	5.865	2.933	0.052	0.81	4–14
Kangaroo 2	1000	8.521	5.681	0.219	0.65	4–15
Kangaroo 3	1000	24.604	2.460	0.441	0.27	2–19
Overall score		3.691				

Table 4: The final submission. The score was reproduced exactly (3.691214) by scoring the file from a clean copy of the course repository in a fresh process.

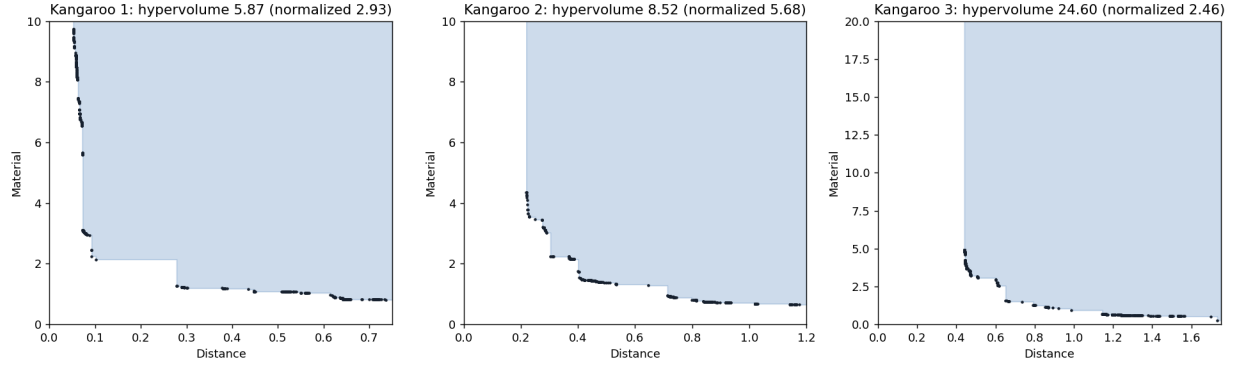


Figure 4: Final Pareto fronts inside their limit boxes.

Figures 5 to 7 show, for each target, the most accurate design, the best-balanced design (the one that alone dominates the largest area) and the lightest design.

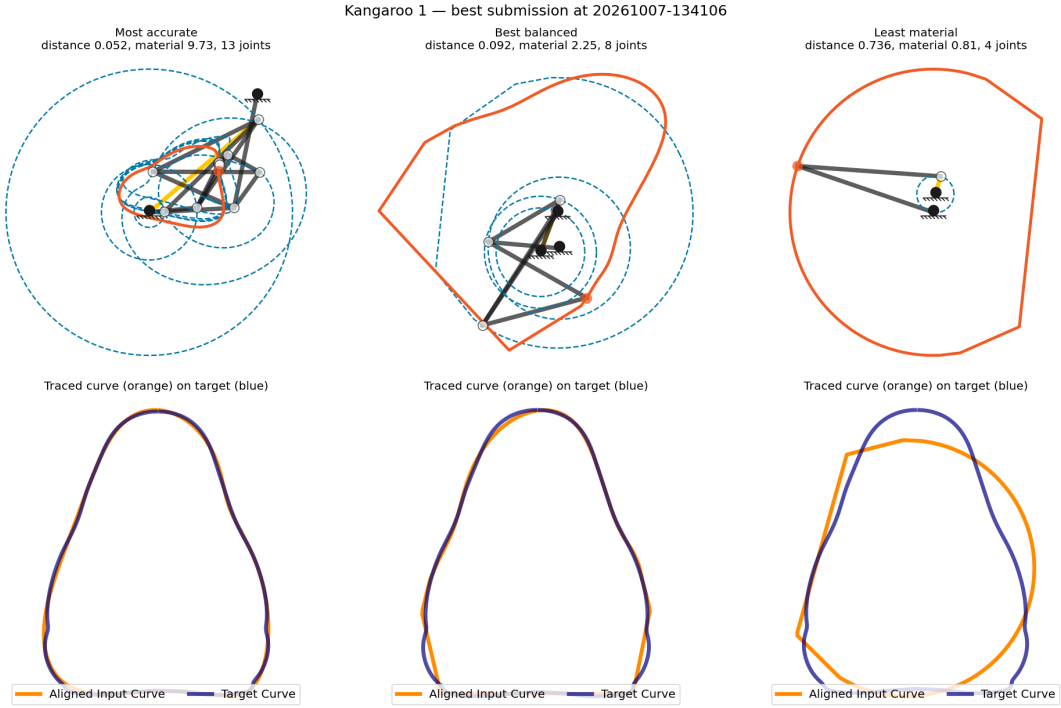


Figure 5: Kangaroo 1: mechanisms (top) and traced curves on the target (bottom).

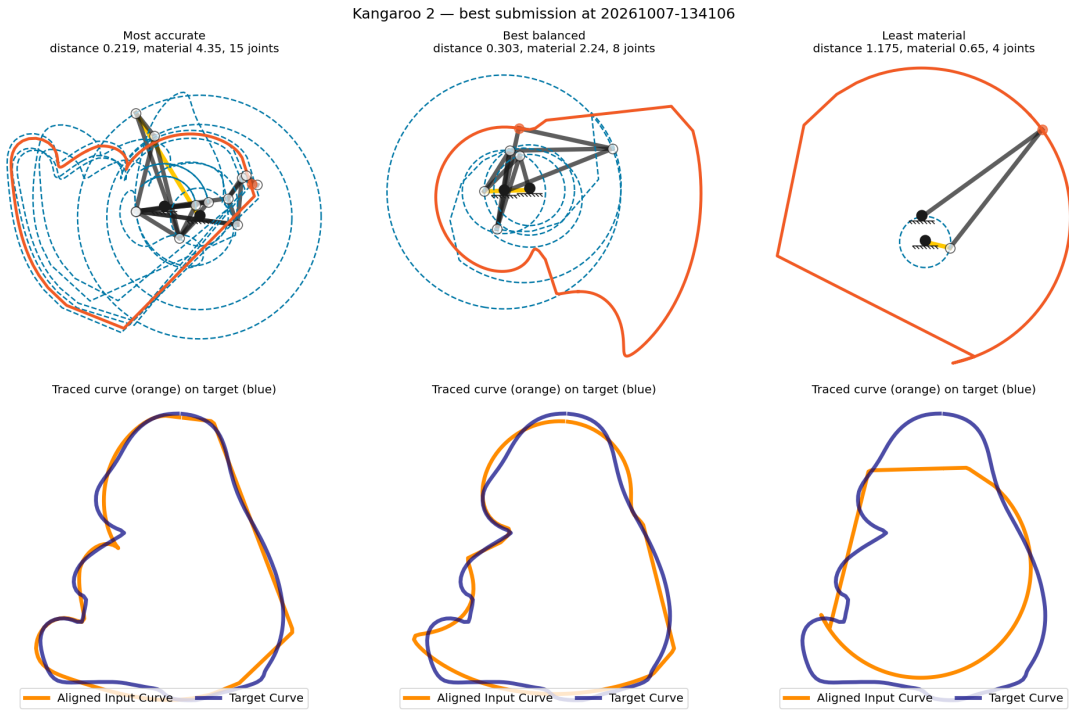


Figure 6: Kangaroo 2.

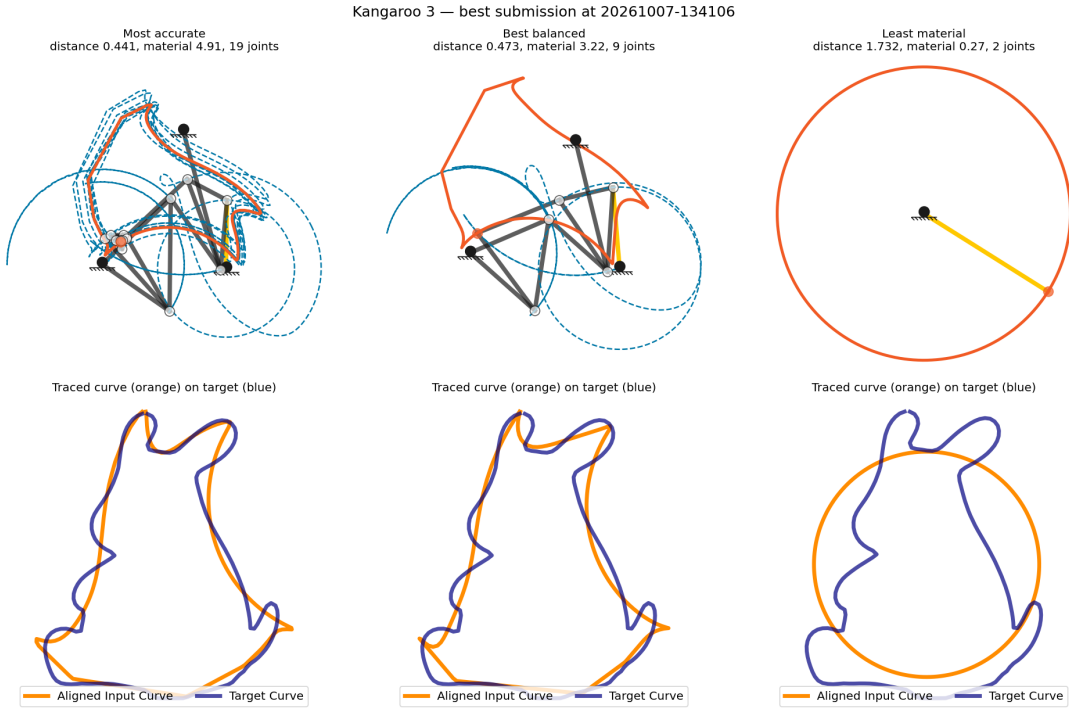


Figure 7: Kangaroo 3. The lightest design is a bare crank tracing a circle.

5 Discussion

Where the score came from. Random screening alone reached 2.78 of the final 3.69, and the first refinement added 0.65 more. Everything after that added 0.26 in total. Of the later stages, the wide refinement added the most (0.17), and it did so by making designs cheaper, not more accurate.

Accuracy against material. For Kangaroo 1 the most accurate and best-balanced designs are close in distance but far apart in material: the last 0.04 of distance costs more than four times the links. Kangaroo 2 shows the same pattern more mildly. Kangaroo 3 is different: its most accurate and best-balanced designs are close in both, and its front stops near a distance of 0.44 however much material is allowed. The material limit is not binding there: the most accurate design uses 4.91 of the 20 allowed. It does use 19 of the 20 joints allowed, but the ten joints it has beyond the best-balanced design lower the distance only from 0.473 to 0.441, so we see no sign that a higher joint limit alone would capture the missing detail. What is missing is a link layout that traces the tail, and our search did not find one.

The bare crank. The lightest Kangaroo 3 design is a two-joint crank whose circle is just inside the distance limit. The course scorer accepts it, but the submission page states 4 to 20 joints per mechanism, so it may not be counted. It is worth 0.005 of Kangaroo 3's hypervolume.

What we would try next. Since joint positions are exhausted, the next step would be a structural search aimed at the tail: growth moves scored by the error in the 55 to 65% stretch of the perimeter instead of the total distance, or a screen for sub-mechanisms that trace a narrow spike, attached to the current best body.

6 Compute and reproducibility

Everything ran as CPU-only Slurm jobs on MIT ORCD Engaging, mostly on the preemptable partition, where a job can be killed without warning. Every script therefore checkpoints about once a minute and resumes exactly where it stopped. Each stage merges its designs with the current best submission per problem, re-scores with the course scorer, and keeps the result only if it is higher. The file `CODE.md` in the code package describes each script and gives the commands that reproduce the pipeline.